Online Quiz Application

script.js

```
// ----- Question bank (one category shown) -----
const bank = {
  ai: [
    { q: "What does AI stand for?", o: ["Automated Input", "Artificial Intelligence", "Applied Internet", "A
    { q: "Which of these is a type of machine learning?", o: ["Supervised learning", "Compiled learning", "S
    { q: "What is a dataset used for in machine learning?", o: ["Styling a web page", "Training and testing
    { q: "Which language is most popular for AI programming?", o: ["HTML", "CSS", "Python", "SQL"], a: 2 },
    { q: "ChatGPT and Claude are examples of what?", o: ["Search engines", "Operating systems", "Large langu
    { q: "What is a neural network inspired by?", o: ["The human brain", "A computer fan", "Electric wires",
  ]
  // web: [...], python: [...] (same format)
};

const TIME_PER_Q = 15;

function shuffle(arr) {
  const a = [...arr];
  for (let i = a.length - 1; i > 0; i--) {
    const j = Math.floor(Math.random() * (i + 1));
    [a[i], a[j]] = [a[j], a[i]];
  }
  return a;
}

el("startBtn").addEventListener("click", () => {
  const name = el("username").value.trim();
  if (name.length < 2) {
    el("msg").textContent = "Please enter a name with at least 2 characters.";
    return;
  }
  el("msg").textContent = "";
  userName = name;
  startQuiz();
});

// ---------- Quiz logic ----------
function startQuiz() {
  // shuffle questions and also shuffle options of each question
  questions = shuffle(bank[category]).map(item => {
    const correctText = item.o[item.a];
    const opts = shuffle(item.o);
    return { q: item.q, o: opts, a: opts.indexOf(correctText) };
  });
  index = 0;
  score = 0;
  history = [];
  showScreen("quizScreen");
  loadQuestion();
}

function startTimer() {
  clearInterval(timer);
  timeLeft = TIME_PER_Q;
  paintTimer();
  timer = setInterval(() => {
    timeLeft--;
    paintTimer();
    if (timeLeft <= 0) answer(-1);
  }, 1000);
}

function answer(choice) {
```

```
  if (locked) return;
  locked = true;
  clearInterval(timer);

  const item = questions[index];
  const opts = document.querySelectorAll(".opt");
  opts.forEach((b, i) => {
    b.disabled = true;
    if (i === item.a) b.classList.add("right");
    if (i === choice && choice !== item.a) b.classList.add("wrong");
  });

  let status;
  if (choice === -1) status = "timeout";
  else if (choice === item.a) { status = "correct"; score++; }
  else status = "wrong";

  history.push({
    q: item.q,
    chosen: choice === -1 ? "No answer" : item.o[choice],
    correct: item.o[item.a],
    status
  });

  el("scoreLive").textContent = "Score: " + score;
  el("fiftyBtn").disabled = true;
  el("nextBtn").textContent = index === questions.length - 1 ? "Finish" : "Next";
  el("nextBtn").classList.remove("hidden");
}

// 50-50 lifeline: hide two wrong options (one use per quiz)
el("fiftyBtn").addEventListener("click", () => {
  if (locked || fiftyUsed) return;
  fiftyUsed = true;
  el("fiftyBtn").disabled = true;
  const item = questions[index];
  const wrongIdx = [0, 1, 2, 3].filter(i => i !== item.a);
  shuffle(wrongIdx).slice(0, 2).forEach(i => {
    const b = document.querySelectorAll(".opt")[i];
    b.classList.add("removed");
    b.disabled = true;
  });
});

function showResult() {
  el("fill").style.width = "100%";
  showScreen("resultScreen");

  const total = questions.length;
  const pct = Math.round((score / total) * 100);
  const correct = history.filter(h => h.status === "correct").length;
  const wrong = history.filter(h => h.status === "wrong").length;
  const skipped = history.filter(h => h.status === "timeout").length;

  // save best score for the category
  const best = getBest();
  let newBest = false;
  if (best[category] === undefined || score > best[category]) {
    best[category] = score;
    newBest = true;
    try { localStorage.setItem("aiQuizBest", JSON.stringify(best)); } catch (e) {}
  }
  showBest();

  // ... title, stats and answer review are filled in here
}
```